

Predictive Modelling on the Type of Patients that will More Likely Accept the Service

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1 COMPANY INTRODUCTION

- The Agency for Integrated Care (AIC) coordinates care across healthcare and community providers to enable seamless, person-centred care for Singapore's population.



CRITICAL : FOCUS

- The National Referral System centralizes and matches patients to appropriate care services such as nursing home, centre-based and home care services based on their assessed needs.

2 OBJECTIVES

- Which factors determine whether patients are more likely to accept or reject services
- Whether access to community-based services delay the need for residential care
- Which factors in patients that utilize community care delay the need for residential care

3 METHODOLOGY

1. Data Cleaning

- Removed irrelevant data (dates, diagnosis etc) due to excessive noise, normalized blanks with their equivalent value in the existing data
- Merged the different datasets based on Application ID to create a central hub dataset



2. Feature Selection | Mutual Information, Cramér's V, Random Forest Importance

- Identifies and ranks features by predictive strength, allowing prioritisation of key factors for model training



3. Model Training and Selection

- Trained different Machine Learning Models like XGBoost, Random Forest, Logistic Regression, LightGBM and developed a 2-Level Prediction System:
- Level 1 → Binary Classification of Admit vs Non-Admit
- Level 2 → Further categorizes Non-Admit cases

Version	Test Accuracy	Test Macro F1	Key Changes
Flat 7-class (best)	63.30%	0.318	Baseline – SMOTE + Random Forest
3-class simplification	54.20%	0.469	Collapse to Admit / Withdrawn / Other
V3 -- 3-level hierarchy	64.80%	0.307	3-level tree with CatBoost
V4 – Simplified hierarchy	54.40%	0.431	2-level, 4-class target
V5 – Confidence gating	72.60%	0.446	Probability calibration added
V7 – Tuned CatBoost	58.20%	0.407	Hyperparameter tuning
Final Ensemble	66.90%	0.401	CatBoost + LightGBM L1 ensemble

4. Predictive Approach

- Confidence-gating achieved higher accuracy, but the **Ensemble model** was selected for its **more stable and consistent performance across classes**.



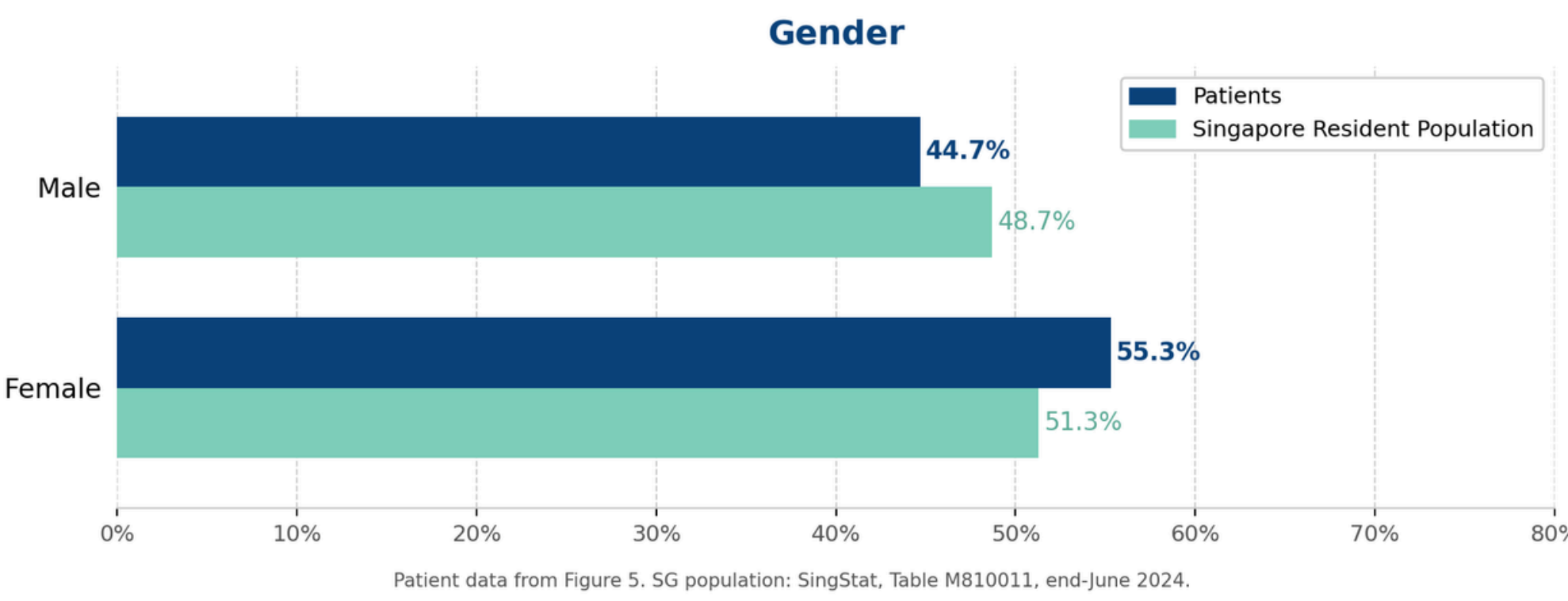
Problem Definition

To develop a predictive model using referral data to analyze patterns, identify key factors and generate actionable insights for effective decision-making

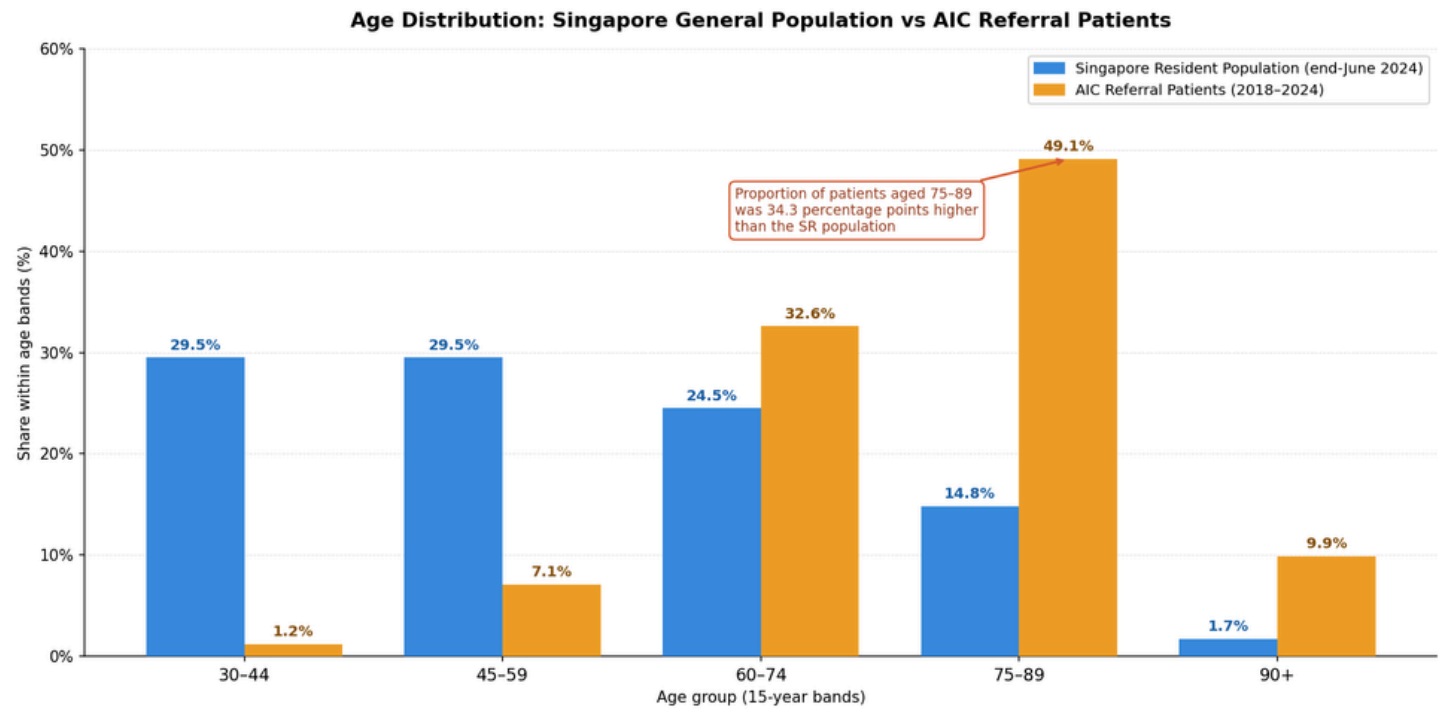
ANALYSIS

*For the purpose of the study, referrals that were withdrawn before assignment to providers were excluded.

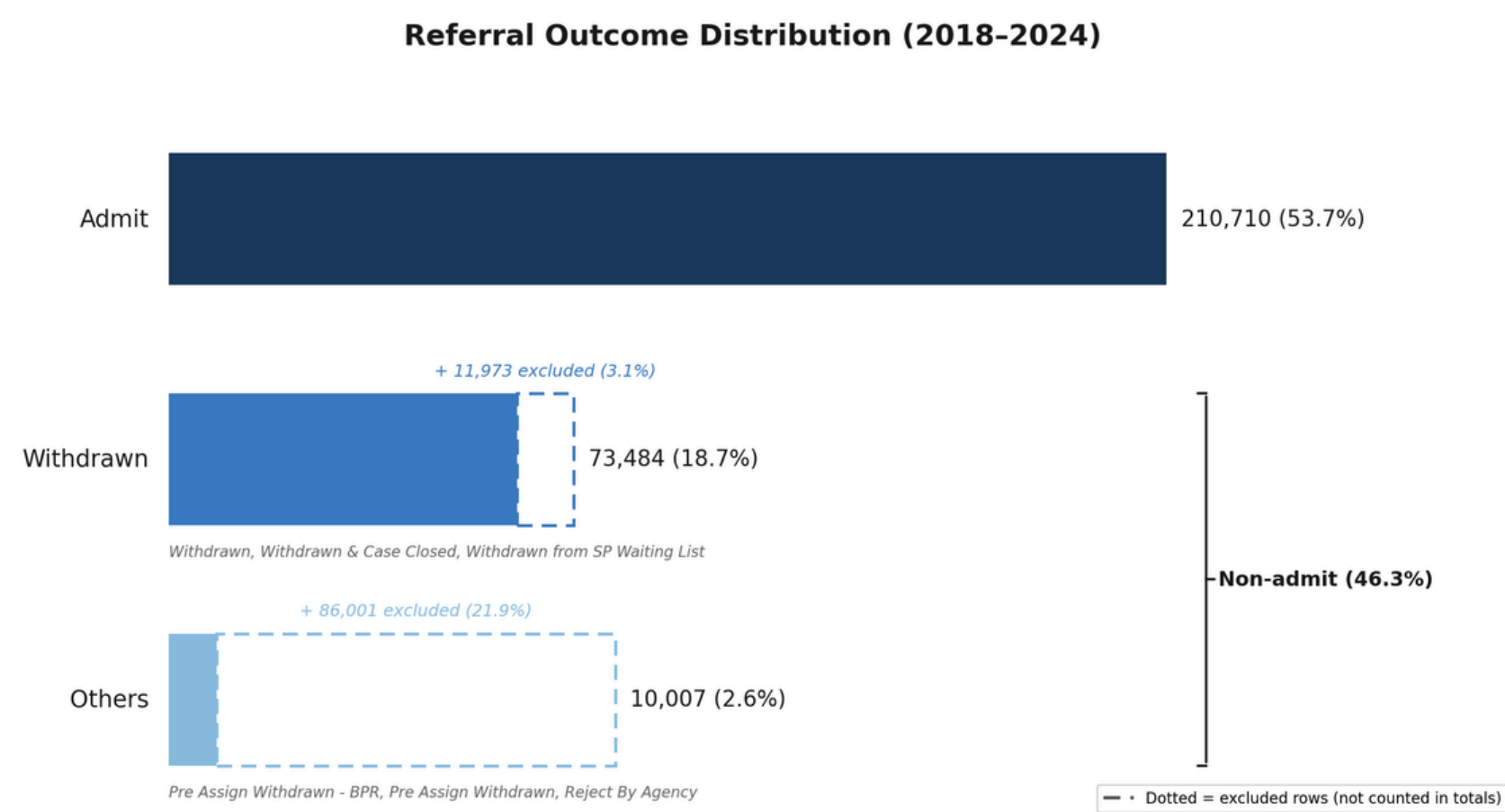
55% of patients were female, slightly above their share of the general population, and **11 percentage points higher** than their male counterpart.



Age 60+ makes up **91.6%** of all AIC referrals, with the **75-89 age group** alone accounting for nearly half (**49.1%**), consistent with AIC's eldercare mission.



53.7% of referrals result in care admission



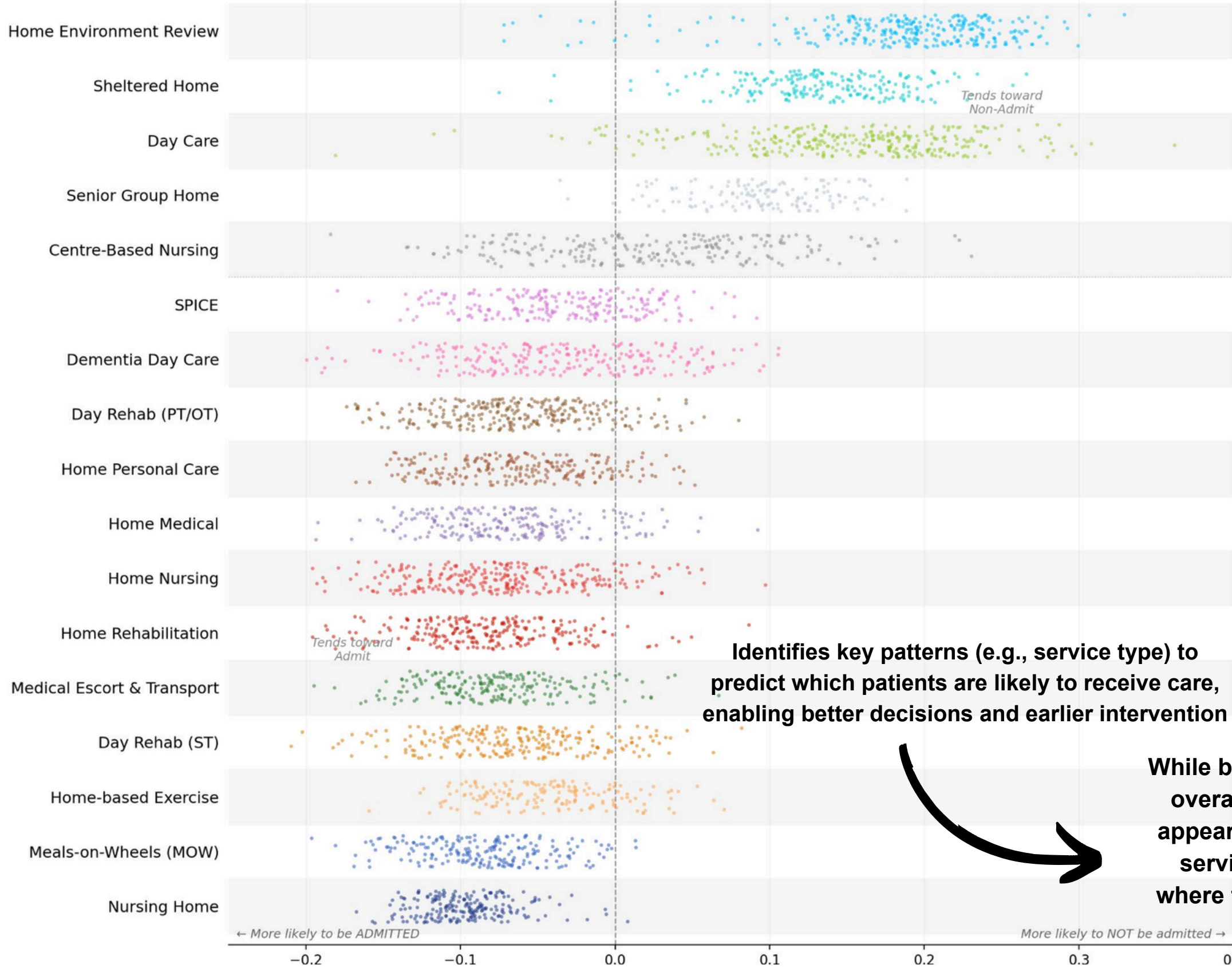
RESULTS

#1 Predictor = Service type

- Right of line - Data points leaning towards non-admission
- Left of line - data points leaning towards admission

Why is this so?

- Social and Bio factors lack a clear distinction between outcomes. (Patients with the same age/living condition/ethnicity/etc are spread across all outcomes)



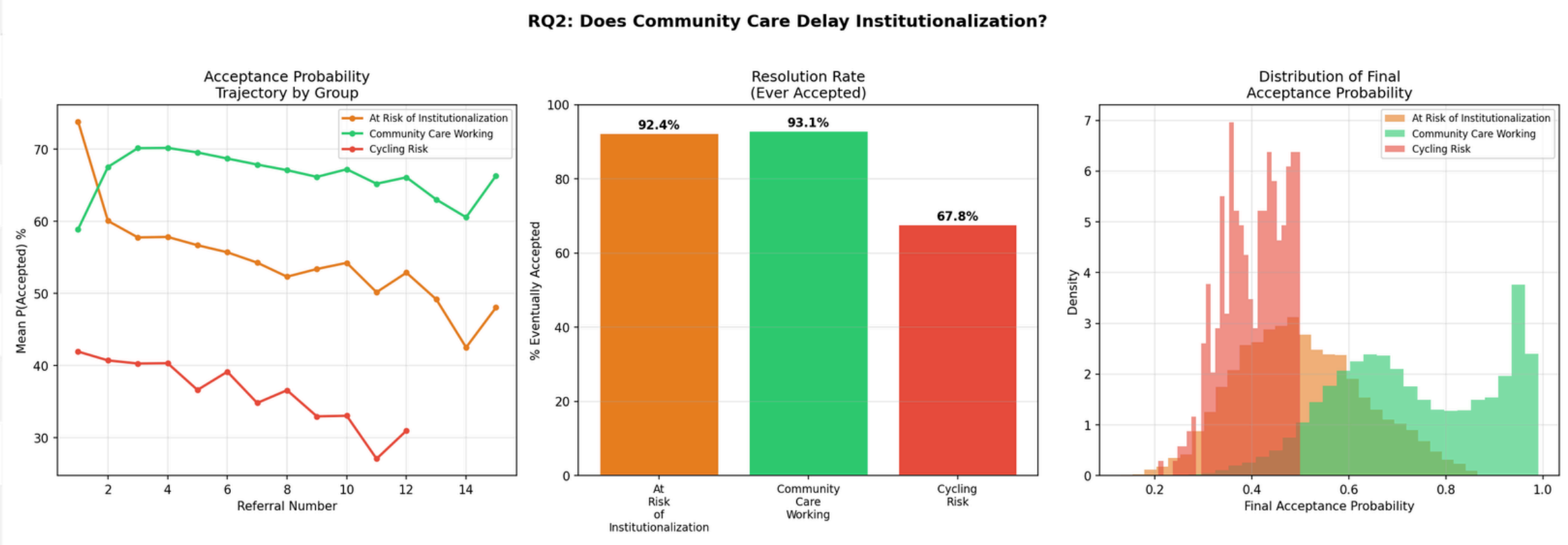
While biosocial factors remain relevant, their overall predictive influence on outcomes appears limited. Disaggregated analysis by service type may help identify contexts where these factors have greater predictive value

Time Prediction Model: Deep Kalman Filter

A model that tracks how patients, that have multiple referrals (3+), their health changes over time and their likelihood of accepting service.

Community Care Outcomes on Cycling Risk

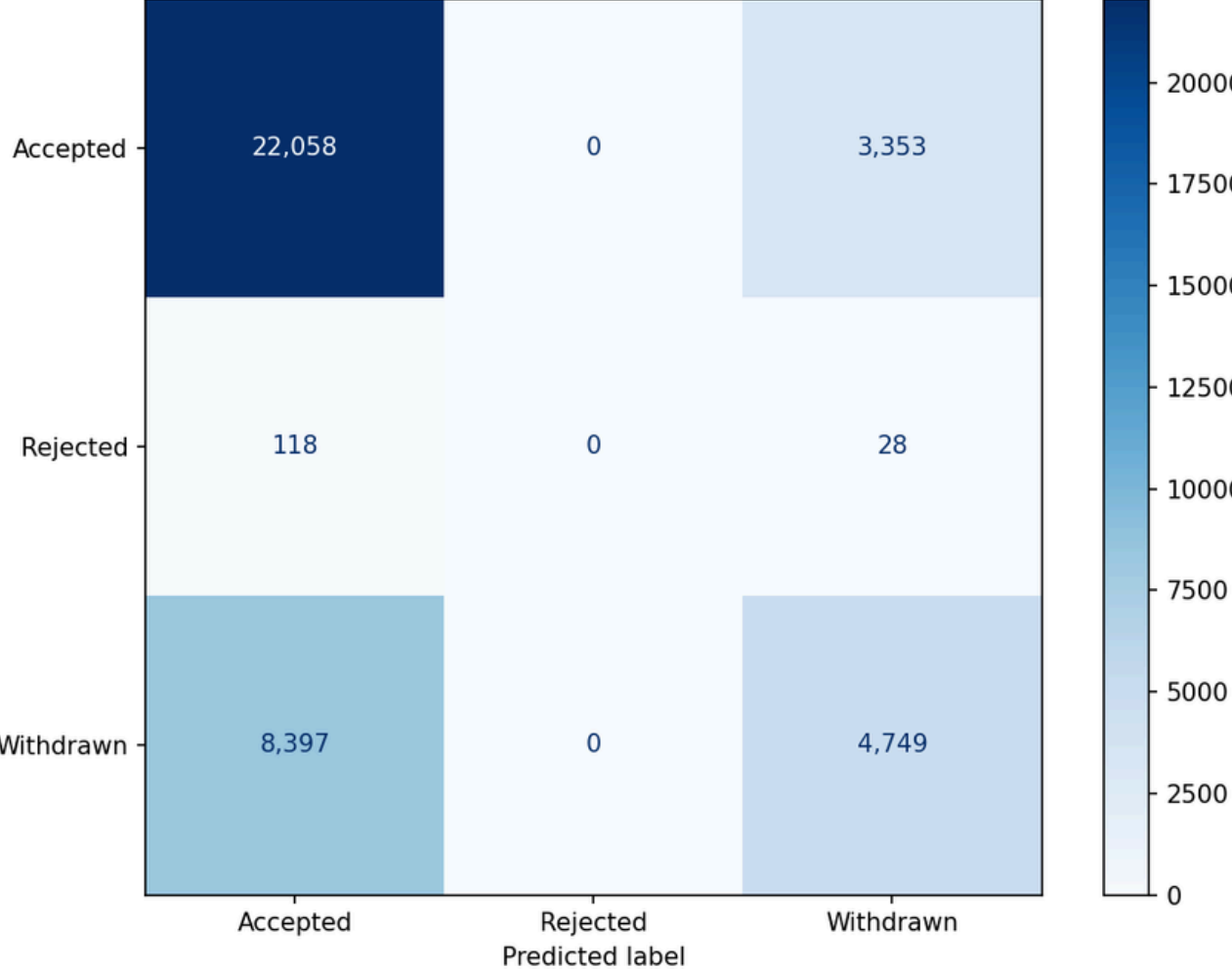
- While non-cycling patient groups see a 92% eventual acceptance rate, Cycling Risk patients face an average withdrawal rate of 32.3%.
- The third and fourth referral attempts represent a key intervention window, allowing care teams to step in and pivot the care plan to suit the patient's new circumstances.



Temporal Predictive Model

- Prediction Accuracy at 69.26%, outperforming baseline by identifying temporal referral patterns
- Formal agency rejections are extremely rare, with non-admittance is mainly driven by withdrawals due to patient's medical condition or caregiving capacity

DKF Confusion Matrix (Overall Accuracy: 69.26%)



CONTRIBUTION AND CONCLUSION

Key Contribution:

- Developed a scalable prediction tool that AIC can continuously update when new referral data is available
- Identified key features and patterns, providing insights for future analytics

Moving forward, AIC can integrate the model and collect longitudinal data to support business capacity planning, with future work focused on further tuning to improve accuracy while maintaining stability.

